2024 Hawai‘i Cesspool Hazard Assessment & Prioritization Tool Report and Appendix: Expansion to the Island of Moloka‘i

Principal Investigator and Author:
Dr. Christopher Shuler

Prepared for:
Hawai‘i Coastal Zone Management Program

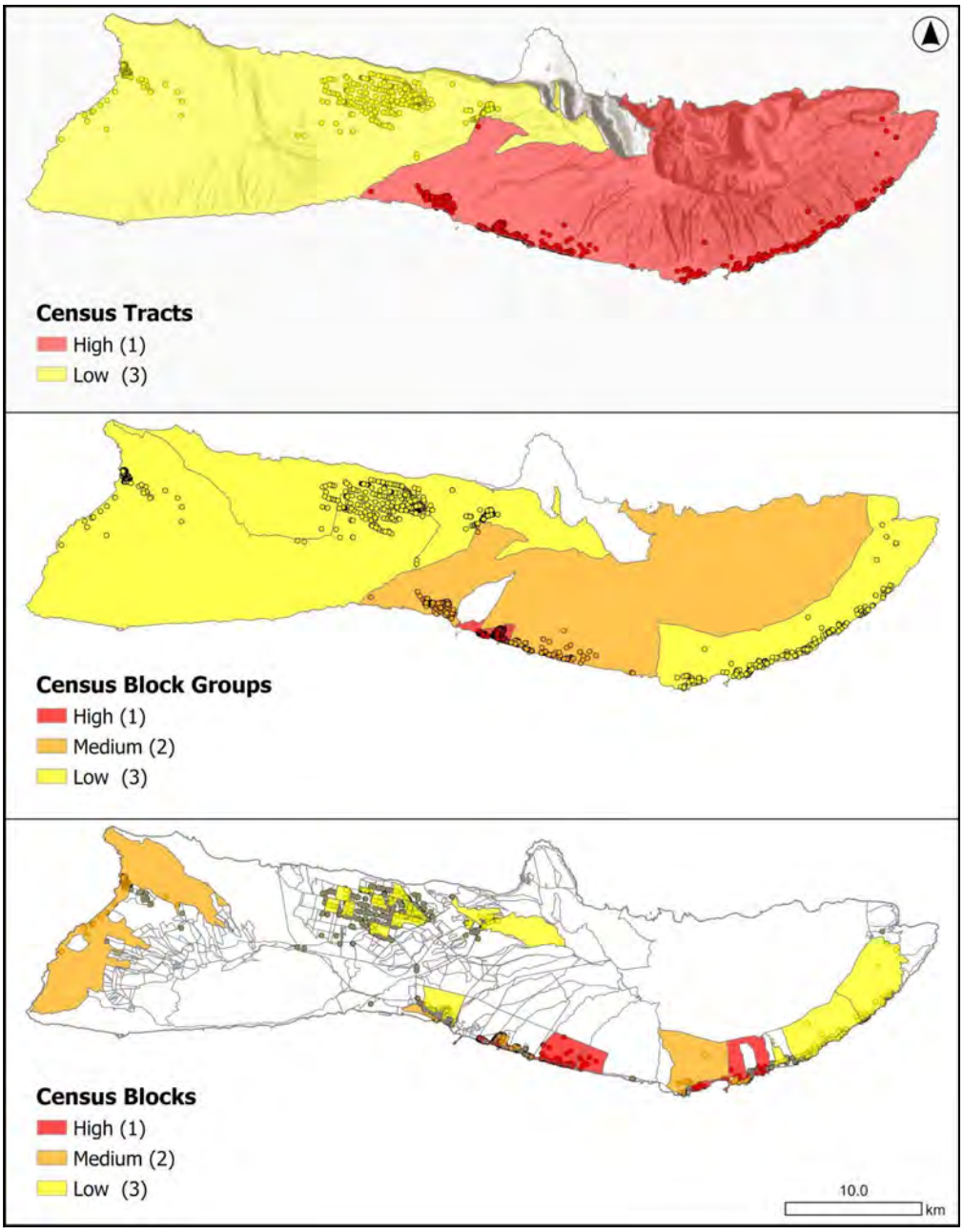


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1. Introduction

This report and appendix to the 2022 Hawai‘i Cesspool Prioritization Tool (HCPT) documents the expansion of the HCPT to include the island of Moloka‘i. Developed as a formal extension of the 2022 HCPT report (Mezzacapo and Shuler, 2022), this work integrates the island of Moloka‘i, and the 1,280 cesspools located therein, into the statewide cesspool hazard framework, and produces output layers using the same geospatial methodology, prioritization schema, and risk factor weighting established in the original project. The initial HCPT assessment, initiated with the 2021 version and updated in 2022, prioritized cesspool replacement efforts for the four main Hawaiian Islands, O‘ahu, Maui, Kaua‘i, and Hawai‘i Island. At that time, Moloka‘i was excluded from the analysis due to significant data gaps, including the absence of a calibrated groundwater model, missing ecosystem risk layers, and an incomplete cesspool inventory. These gaps precluded its inclusion without compromising the statewide consistency of the tool.

Since then, new data products and stakeholder support have made it feasible to develop and apply the HCPT framework to Moloka‘i. Incorporating the island into the tool addresses a key equity gap and ensures that Moloka‘i residents are represented in statewide cesspool hazard assessments and are eligible for future funding and planning programs. This work enhances the utility of the HCPT as a decision-support tool for the Hawai‘i Department of Health, the former Cesspool Conversion Working Group members, and other state agencies tasked with implementing Act 125 and related initiatives. The methodology used in this expansion mirrors that of the original HCPT. It includes the use of updated groundwater flowpaths, refined cesspool location data, and consistent application of fifteen geospatial risk factors to assign prioritization scores. This report and appendix outlines the data acquisition and processing approach specific to Moloka‘i, presents the resulting hazard prioritization scores, and discusses relevant limitations and next steps.

1.1 Objectives

The main objectives of this project are to:

1. Identify and generate the missing datasets needed to assess cesspool hazards on Molokaʻi, including groundwater flowpaths, cesspool inventory, and environmental risk layers.
2. Integrate Molokaʻi-based geospatial data into the existing HCPT framework using the same 15 risk factors, scoring algorithms, and weighting structure as the original model.
3. Calculate final prioritization scores for all 1,280 known cesspools on Molokaʻi to support hazard ranking and mapping.
4. Update HCPT project deliverables, including geospatial layers, documentation, and the public website, to incorporate Molokaʻi results.
5. Enable Molokaʻi communities to participate in state-led cesspool conversion programs by ensuring they are represented in statewide hazard assessments.

2. Methods

This analysis applies the same methodology originally developed for the 2021 HCPT to assess cesspool prioritization on the island of Molokaʻi. The Molokaʻi expansion follows the established geospatial framework, using the original set of fifteen risk factors, standardized scoring algorithms, and weighting structure. Where data were previously unavailable for Molokaʻi, such as groundwater flowpaths and certain ecological layers, new datasets were developed or acquired in coordination with state and local partners. These inputs were processed using the same scripting tools and workflows as the original project, ensuring full compatibility with the statewide HCPT system. All source code and relevant datasets are publicly accessible through published Jupyter notebooks located on GitHub at https://github.com/cshuler/HCPT_Molokai

2.1 Cesspool Inventory Updates

Cesspool location data for Molokaʻi were compiled from the Hawaiʻi Department of Health's Individual Wastewater System (IWS) database and augmented using county tax parcel records. The dataset used in this project incorporates 2020 updates from the DOH Wastewater Branch (v3 from the original HCPT), the IWS Database (v4), and dwelling data from all four counties' tax record systems (v5). The version used in this update (Fig 1.) was generated specifically for Molokaʻi and reflects the most current cesspool inventory

available for the island. As in the original HCPT project, the inventory was refined by removing decommissioned units, excluding parcels with no residential structures, and geolocating cesspools as discrete point features. Where parcels contained multiple cesspools, individual units were separated to ensure one point per cesspool. The final dataset includes 1,280 cesspool units on Molokaʻi. These points form the basis for risk factor evaluation and prioritization scoring.

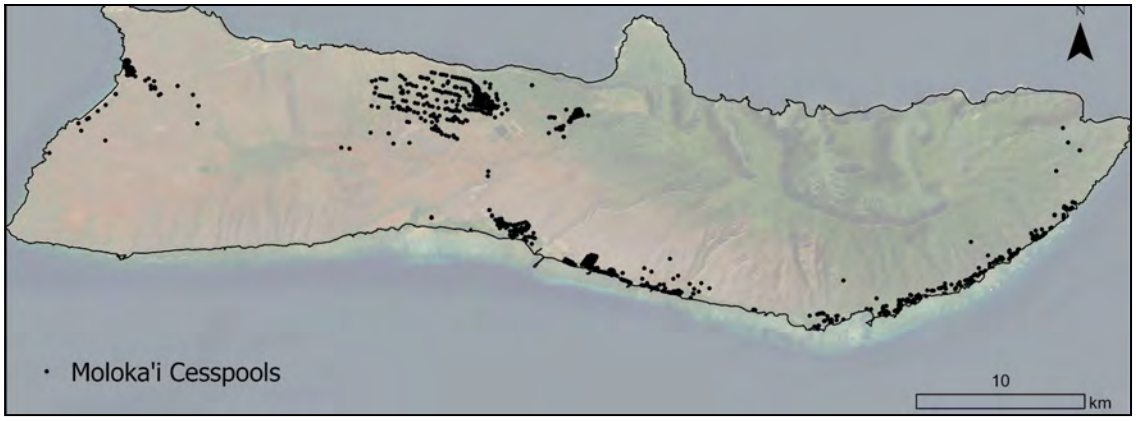


Figure 1: *Distribution of cesspools on Molokaʻi (n = 1,280) based on the updated Hawaiʻi Department of Health IWS database, county tax parcel records, and refined parcel-level geolocation. These point data formed the basis for evaluating risk factors and generating prioritization scores in the Molokaʻi HCPT analysis.*

It should be noted that the currently available cesspool inventory contains a number of known inconsistencies, and the statewide dataset remains in need of further updates. Although an updated inventory is likely to be released soon, it was not available at the time of this report. The HCPT codebase is fully open source, and all components of the prioritization workflow have been designed to accommodate updated OSDS datasets as they become available. Regardless, the inventory used represents the best available data at present, and the actual number of cesspools on Molokaʻi is unlikely to change significantly given the island's largely rural character.

2.2 Groundwater Model and Flowpath Generation

One of the most critical and technically significant tasks in expanding the HCPT to Molokaʻi was generating new groundwater flowpaths (Fig. 2) from a newly acquired, previously calibrated groundwater model provided by the Hawaiʻi Department of Health. The groundwater flowpaths for Molokaʻi cesspools were derived from a MODFLOW-based numerical groundwater model originally developed as part of the Hawaiʻi Source Water Assessment Program (SWAP). This model incorporates detailed hydrogeologic inputs

including recharge estimates, aquifer properties, pumping rates, and boundary conditions specific to the island. Particle tracking to generate both groundwater flowpaths and well capture zones was conducted using MODPATH, which simulates conservative advection of dissolved contaminants by tracing flowpaths from cesspool locations to coastal discharge points. Model calibration was performed using hydraulic head data from wells and springs, and sensitivity analyses were conducted to evaluate the impact of varying parameters such as recharge, porosity, and hydraulic conductivity. Model details are provided in (Whittier et al., 2010).

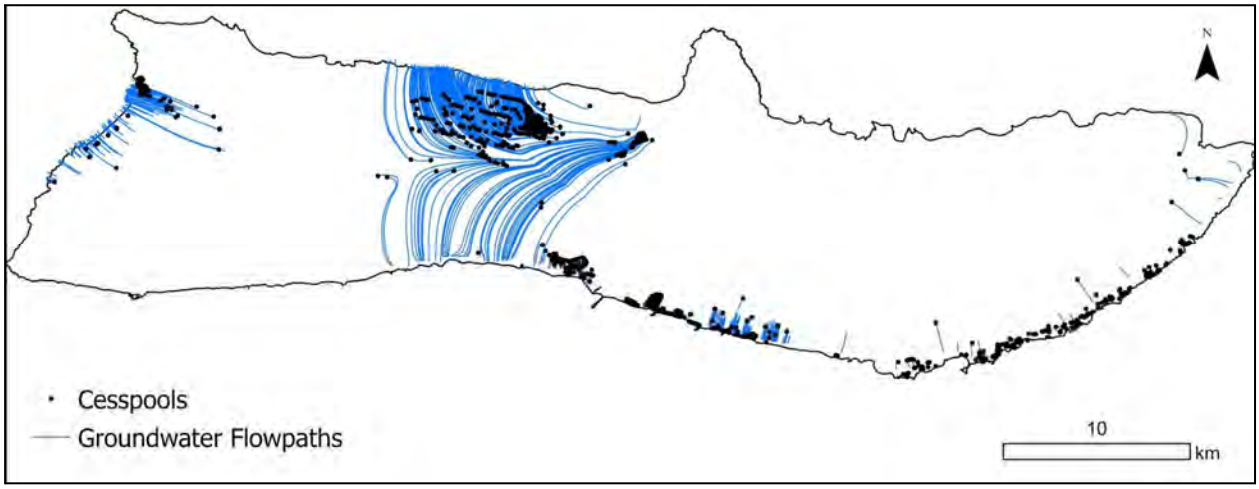


Figure 2. *Groundwater flowpaths for Moloka‘i cesspools, generated using a MODFLOW-based groundwater model and MODPATH particle tracking. Flowpaths connect cesspool sources (black points) to coastal discharge zones (blue lines), providing the linkage necessary to evaluate downstream ecological and human health risk factors.*

Flowpaths were generated using the MODPATH particle-tracking module within the groundwater model. Flowpaths were initiated at cesspool locations and traced downgradient through the saturated zone to the coastline. Endpoints were assigned to evenly spaced shoreline segments represented by 250m square grid cells. This enabled risk factor values, such as coral reef condition, user-days, and ocean circulation, to be mapped back to the originating cesspools. This approach ensures comparability across all islands and allows the same fifteen-factor prioritization schema to be applied to Moloka‘i with full methodological consistency.

2.3 Input Data: Risk Factors

2.3.1 Distance to Coastline, Streams, Wetlands, and Wells

All of the risk factors from the original HCPT model were retained for use in the Molokaʻi expansion. Overall, the data layers, methods, and scoring algorithms remained consistent. Individual risk-factor scores for distance to coastline, distance to streams and wetlands, and distance to municipal and domestic wells were all evaluated by calculating the distance from each cesspool to the respective hydrologic feature using the threshold-decay scoring function as described in the 2022 HCPT report. Threshold distances and data sources are described below.

- For **Coastline Proximity**: the threshold was set at 15.24 meters (50 feet), which is the shoreline setback limit defined in Hawaiʻi Administrative Rules Title 11, Chapter 62 (HAR 11-62). Scores above this threshold are set at the maximum of 100 and decrease with increasing distance from the coastline. A coastline shapefile procured from HI-GIS Online (2020a), was used for the analysis.
- For **Stream and Wetland Proximity**: the threshold was also set at 15.24 meters (50 feet), consistent with the regulatory setback defined in HAR 11-62. Scores above this threshold are assigned the maximum value of 100 and decrease with increasing distance from the stream or wetland feature (HI-GIS Online, 2021, 2020b).
- For **Proximity to Municipal and Domestic Drinking Water Wells**: The threshold distance for municipal wells was set at 17 meters (56 feet), based on guidance from the Hawaiʻi Wellhead Protection Program, which defines this distance as the Zone A radius for protecting against direct contamination near well casings (Whittier et al., 2010). For domestic wells, a more conservative threshold of 28 feet (8.5 meters) was used to reflect their typically shallower construction and closer proximity to cesspool units. Locations of municipal and domestic drinking water wells were obtained through a data request to the Hawaiʻi Commission on Water Resource Management (CWRM) based on the State Well Inventory Database.
- To evaluate whether cesspools were located inside of **Well Capture Zones,** a simple binary classifier was used to classify each cesspool as either inside or outside of a designated capture area. Specifically, cesspools located within the 2-year travel time zone (Zone B) received a score of 100, those within the 10-year zone (Zone C) received a score of 50, and all others received a score of 0. Well capture zones were generated using the Molokaʻi groundwater model provided by the Hawaiʻi Department of Health and were defined through reverse particle tracking using MODPATH. Note that well location and capture zone data in Hawaiʻi is somewhat sensitive, and though it is available upon request to the Hawaiʻi Department of

Health or Commission on Water Resources Management it is not publishable, therefore the code-base and figures do not directly contain these data (Fig. 3).

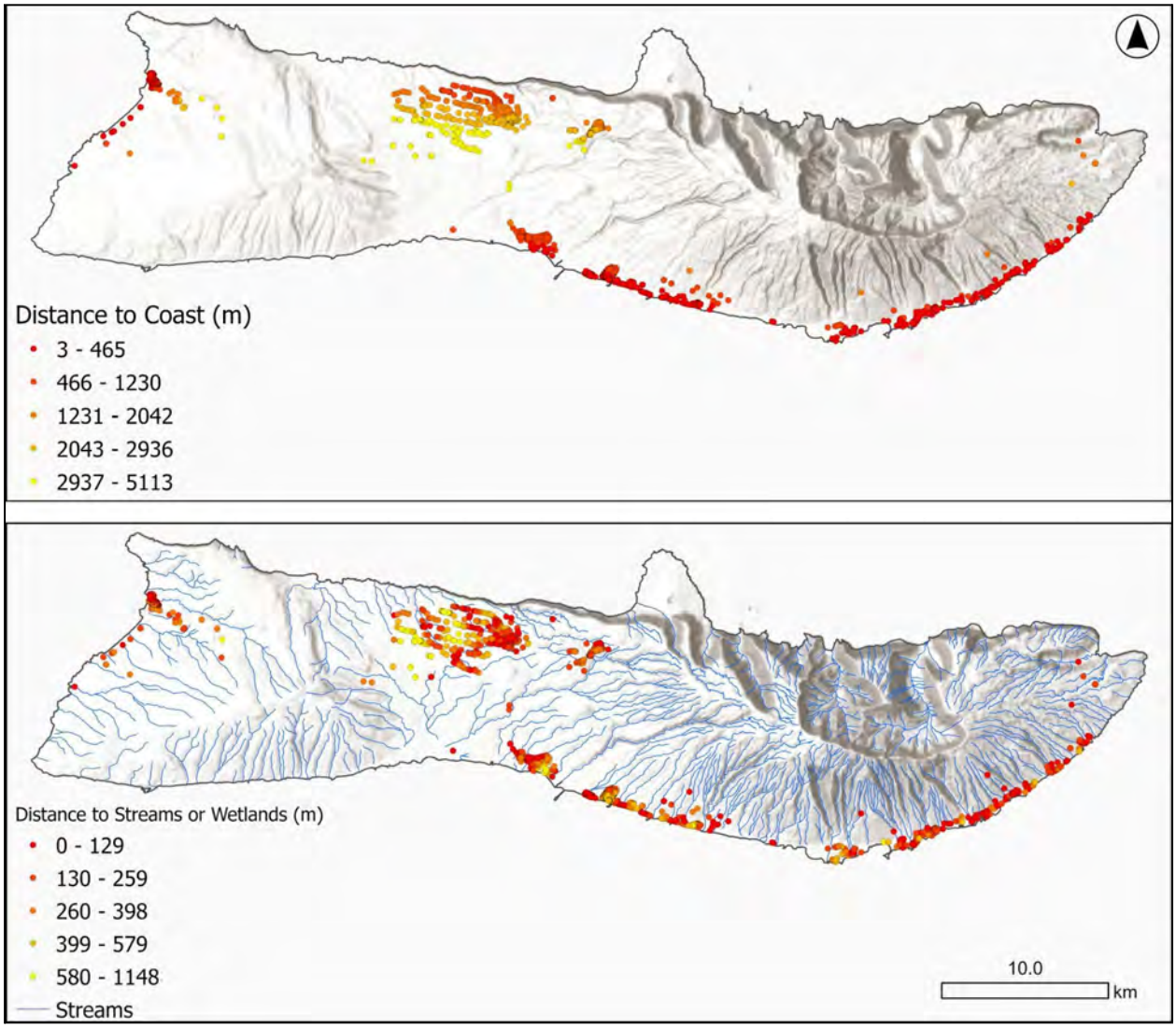


Figure 3. *Distance-based risk factor inputs for Moloka'i cesspools. Top: distance to coastline (meters) calculated from the state shoreline dataset. Bottom: distance to streams and wetlands (meters) from state hydrography layers.*

2.3.2 Water Table Elevation (Depth to Groundwater)

The depth from land surface to the water table was retained as a risk factor in the Molokaʻi expansion without modification to methodology. As in the original HCPT model, this factor reflects the reduced potential for effluent attenuation in areas where cesspools are installed in shallow, saturated soils. For Molokaʻi, a gridded depth-to-water surface was generated using output from the MODFLOW-based groundwater model developed by the Hawaiʻi Department of Health. Each cesspool was assigned a depth-to-groundwater value by intersecting its location with this gridded surface. Depth to groundwater was calculated as the difference between land surface elevation and the water table at each cesspool location (Fig. 4). Elevations were derived from a statewide <1-meter LiDAR DEM (HI-GIS Online, 2017), supplemented with 10-meter DEM data (USGS, 2013) where gaps existed. Risk scores were then calculated using the same threshold-decay function and thresholds established in the 2022 HCPT report.

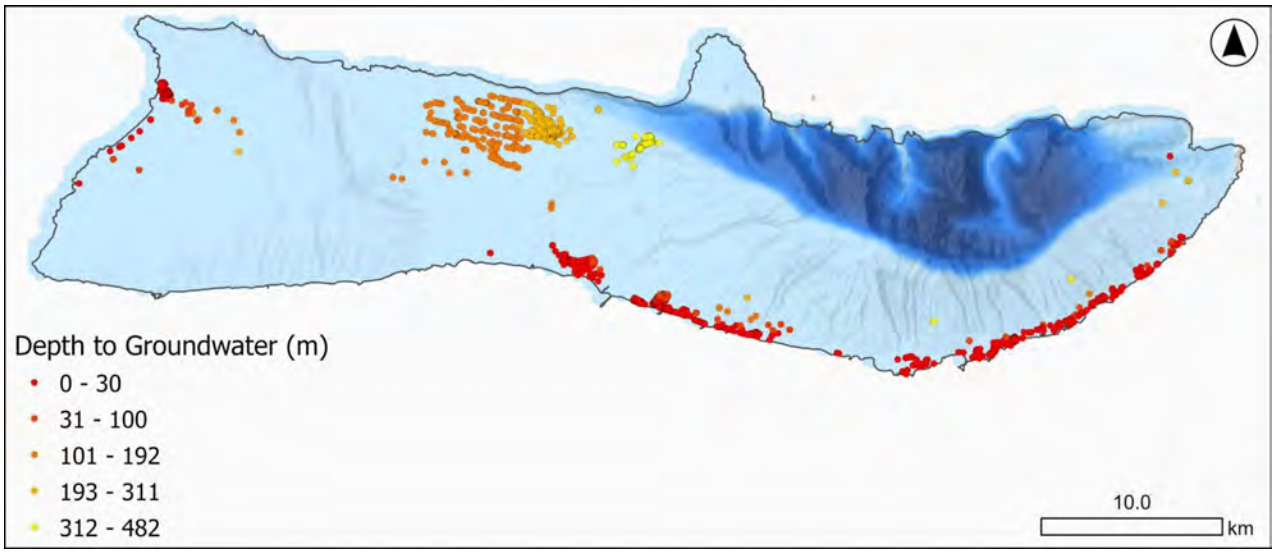


Figure 4. *Depth to groundwater risk factor for Molokaʻi cesspools, derived from the DOH MODFLOW groundwater model and statewide elevation data. Points are symbolized by depth-to-water intervals (meters), with shallower values indicating higher risk of effluent reaching the saturated zone.*

2.3.3 Sea Level Rise (SLR)

The same approach used in the original HCPT model was applied to evaluate cesspool vulnerability to sea level rise on Molokaʻi. However, new updated SLR data were obtained on April 29, 2025 from the latest update of the Hawaiʻi SLR viewer (Hawaiʻi Climate Change Mitigation and Adaptation Commission, 2021) . Risk scores were assigned based on both

horizontal surface flooding and vertical groundwater inundation, reflecting the dual mechanisms by which sea level rise can impair cesspool function. Newer data is indexed to an amount of SLR (e.g. 3.2 ft), rather than a projection date (e.g. 2100) to better reflect uncertainties in emissions scenarios.

Horizontal inundation zones were obtained directly from the Hawai‘i Sea Level Rise Viewer (Tetra Tech, Inc. and University of Hawai‘i Coastal Geology Group, 2017), which provides modeled flood extents for three future scenarios: 0.5 feet (formerly 2030), 1.1 feet (formerly 2050), and 3.2 feet (formerly 2100). In addition to surface flooding, vertical inundation of the subsurface was considered by analyzing reductions in the unsaturated zone. Following the HCPT methodology, each cesspool's depth to groundwater was recalculated after applying a sea level rise offset. If the resulting depth to groundwater fell below 4.4 meters (14.4 feet), the cesspool was classified as vertically inundated. Cesspools that were either located within the horizontal sea level rise inundation zones or that met the condition of vertical inundation for 0.5 ft, 1.1 ft, or 3.2 ft, were assigned risk scores of 100, 66, and 33, respectively (Fig. 5).

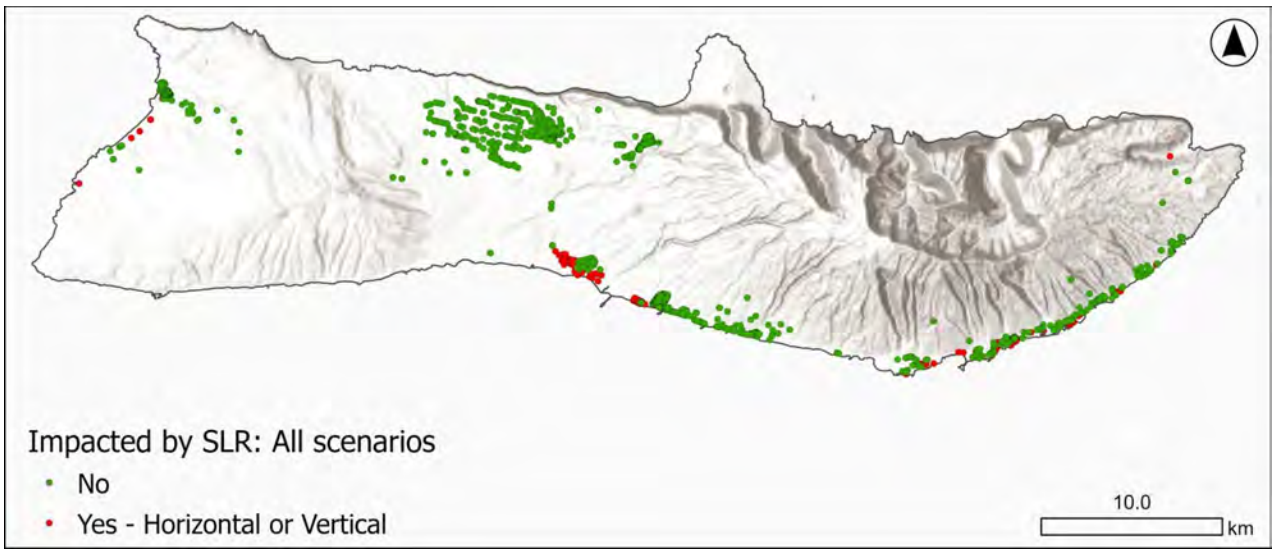


Figure 5. *Cesspools on Moloka‘i identified as vulnerable to sea level rise (SLR) impacts under all modeled scenarios: 0.5 ft, 1.1 ft, and 3.2 ft of SLR. Red points indicate units affected by either horizontal surface inundation or vertical groundwater inundation (unsaturated zone reduced below 4.4 m). Green points indicate cesspools not impacted by SLR in any scenario.*

2.3.4 Soil Characteristics

Soil drainage properties influence how quickly effluent percolates through the subsurface and whether it is likely to reach groundwater or surface water bodies. As in the original HCPT model, soils were classified into drainage categories based on NRCS SSURGO data, and risk scores were assigned according to the mapped soil drainage class at each cesspool location. For Molokaʻi, no modifications to the scoring approach were required. The NRCS SSURGO soil database already includes detailed spatial coverage for Molokaʻi. Each cesspool was intersected with the soil dataset and assigned a risk score (Fig. 6) based on its mapped drainage classification, following the same scale and weights as the original HCPT implementation.

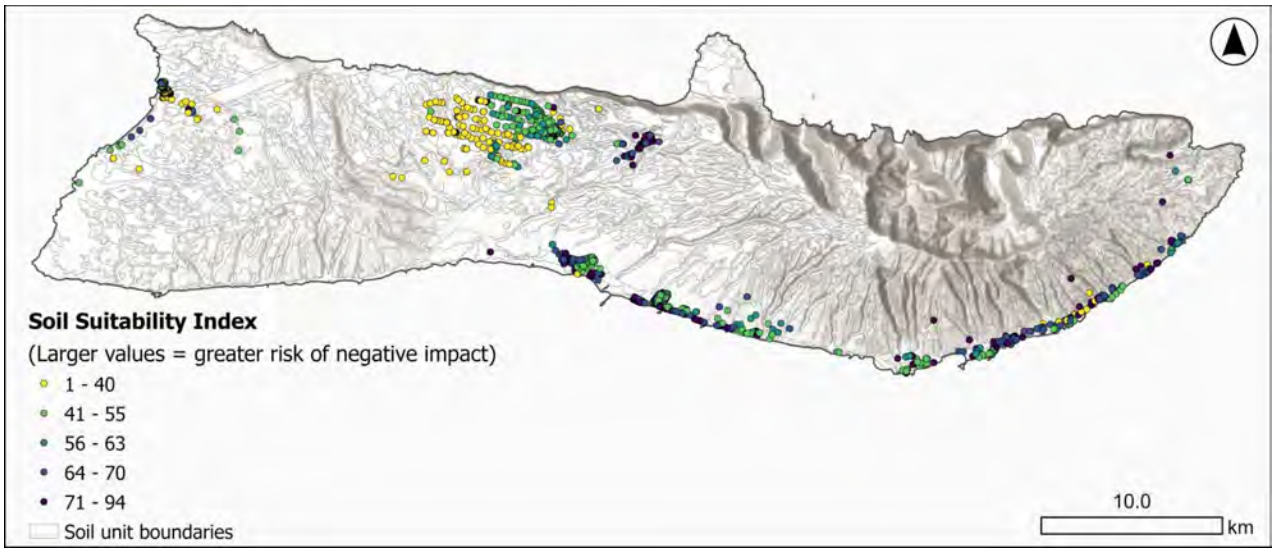


Figure 6. *Soil suitability index for cesspools on Molokaʻi. Larger values and darker colors indicate units in soils that are less suitable for attenuating wastewater contaminants.*

2.3.5 Cesspool density

Cesspool density was retained as a risk factor in the Molokaʻi expansion, using the same approach as in the original HCPT model (Fig. 7). Density is defined as the number of cesspools per acre, calculated by aggregating cesspool counts within census block groups and normalizing by land area. This metric provides a relative indication of areas where cumulative loading from multiple cesspools may increase the risk of groundwater or surface water contamination. Consistent with the 2022 HCPT report, the density calculation uses a threshold-based scoring function, with higher scores assigned to areas exceeding the EPA-referenced value of approximately 40 systems per square mile (15.4/km²) (Flanagan et al., 2019). Although this value originates from studies outside Hawaiʻi and cesspool impacts are strongly site-dependent, it remains a useful benchmark for comparing relative density across the island. The approach does not attempt to

establish locally derived density thresholds, which would require site-specific hydrogeologic studies, but instead applies a consistent calculation across census block groups to support prioritization and planning.

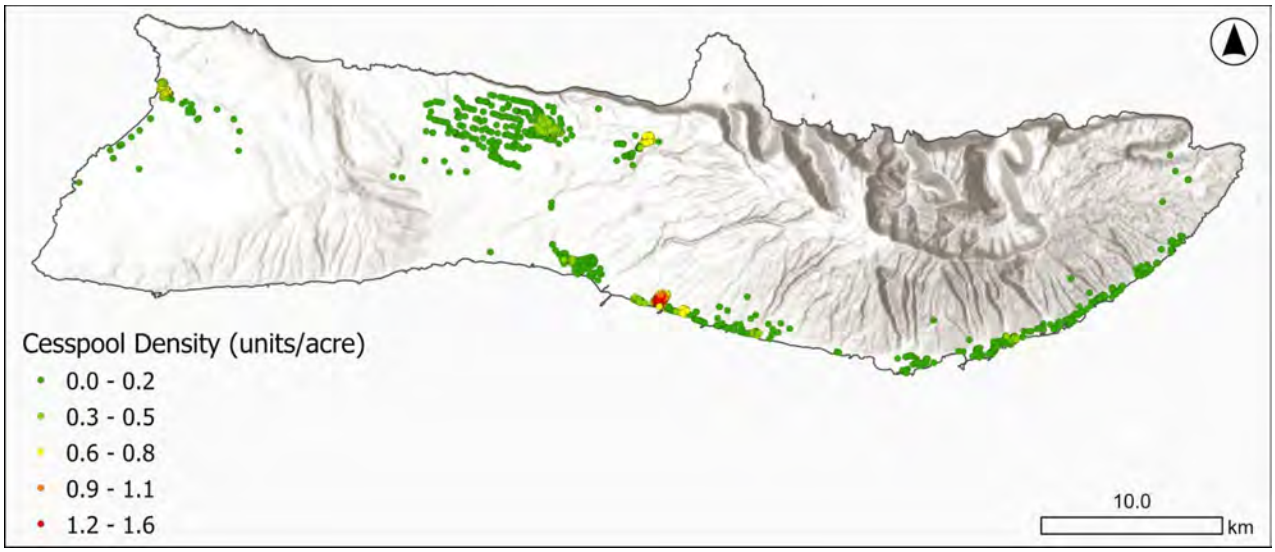


Figure 7. *Cesspool density on Molokaʻi expressed as units per acre. Warmer colors indicate higher densities, with localized hotspots evident in Kaunakakai and parts of the south shore, while most other areas show dispersed, lower-density patterns.*

2.3.6 Precipitation

Precipitation (rainfall) was included as a risk factor for the Molokaʻi expansion following the original HCPT approach. While rainfall does not directly influence the suitability of cesspool placement, it serves as a proxy for groundwater recharge, which controls the dilution of wastewater effluent once it enters the subsurface. Areas with higher rainfall are assumed to provide greater recharge and therefore greater dilution capacity, reducing the relative concentration and impact of contaminants. Average annual rainfall values for Molokaʻi were extracted from the statewide gridded rainfall dataset developed by Giambelluca et al. (2013). Each cesspool was assigned a rainfall value by intersecting its location with the rainfall grid (Fig. 8). Consistent with the 2022 HCPT methodology, a threshold-decay function was applied to calculate risk scores, with the threshold set at 0.2 meters per year (8 inches), the lowest rainfall value in the dataset. Cesspools located in areas with higher rainfall received lower risk scores, while those in drier regions received higher scores, reflecting the reduced potential for effluent dilution.

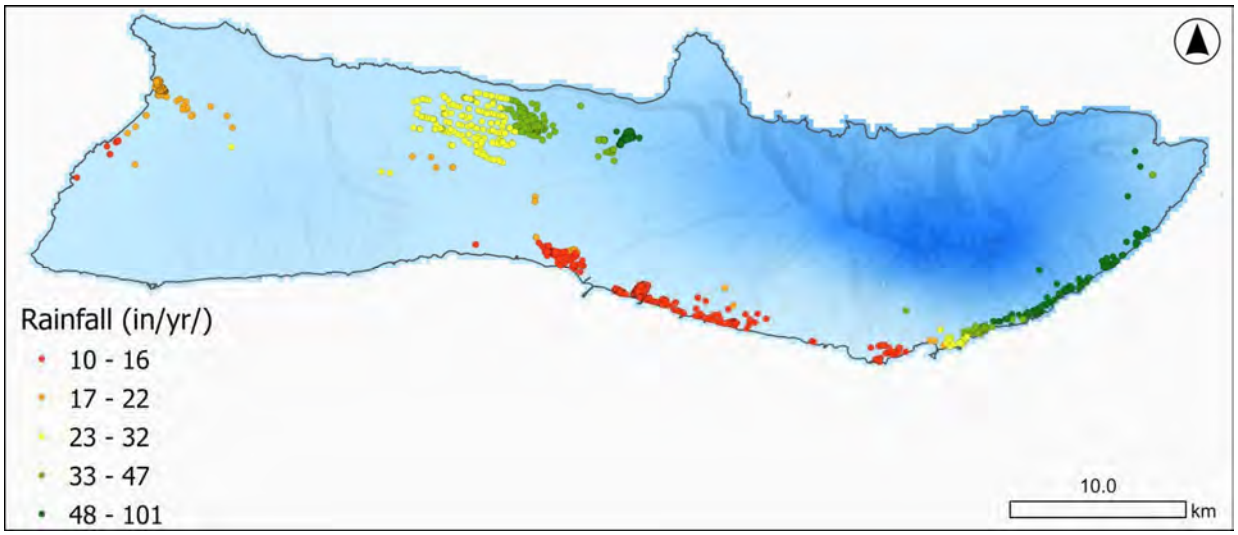


Figure 8. *Average annual rainfall across Moloka‘i, with cesspool locations symbolized by the rainfall values (inches per year) at each site. Lower-rainfall areas along the south shore (reds and oranges) indicate reduced dilution potential, while higher-rainfall upland and eastern areas (greens and blues) reflect greater groundwater recharge capacity.*

2.3.7 Coastal Impact Factors: Coral, Fish, Coastal Use, and Ocean Exposure

The following risk factors represent coastal impacts and are associated with coastline grid cells that are linked to individual cesspools through modeled groundwater flowpaths. Methods from the 2022 HCPT were retained for consistency, and Moloka‘i-specific datasets were updated and generated for this report. For each factor, scores from the relevant coastline grid cell are assigned to the cesspool whose flowpath terminates at that shoreline segment.

- For **Coral cover.** A new coral cover dataset was generated for Moloka‘i as the original datasets from 2021 only covered the four most populated Hawaiian islands. Two coral datasets , 1) current live coral cover and 2) baseline (pre-2015) coral cover were summarized within 0–15 m depth and aggregated to 1 km shoreline segments. Each dataset was then divided into four percentile ranks (e.g., very high, high, moderate, low). To create a single coral cover score for each shoreline segment, the two ranks for each segment were compared and the segment was assigned the higher of the two ranks if they differed. Ranks were converted to 0–100 scores using the scores of 100, 66, 33, and 0 for very high, high, moderate, low coral value respectively. These scores were assigned to cesspools via their coastal discharge cells.
- For **Resource fish biomass and recovery potential.** As with the coral cover dataset a new dataset was generated for this update. Standing biomass of resource fishes and

the theoretical recovery potential if OSDS effluent were removed were each summarized within 0–15 m depth and aggregated to 1 km shoreline segments. Datasets were ranked into four percentile ranks, combined by taking the higher priority rank as above, and translated to 0–100 scores by mapping the scores of 100, 66, 33, and 0 to the priority ranks of very high, high, moderate, low, respectively. Scores were linked to cesspools through the same groundwater flowpath to coastline grid cell linkage (Fig. 9).

- For **Coastline visitation** (user-days). Relative coastal use was derived from geotagged photo data following the established HCPT methodology, with user-days calculated along the shoreline sampling points and mapped to the coastline grid. Because usage is highly skewed, a threshold-decay function was applied with a threshold of 100 user-days to identify high-use segments, and with decreasing scores assigned to areas with lower visitation.
- For **Swimming Beaches.** Unlike other islands, Molokaʻi has no lifeguard towers to delineate designated swimming areas. To identify beaches where people are most likely to enter the water, a shapefile was generated using information from publicly available tourism resources. Specifically, the official Hawaiʻi Tourism Authority website (https://www.gohawaii.com/islands/molokai/things-to-do/beaches) and Visit Molokaʻi (https://visitmolokai.com/wp/molokai-beaches/) were consulted to compile a list of recommended swimming beaches. A total of seven (7) swimming beaches were identified, and coastline segments within 500 meters of these beaches were classified as swimming areas. Cesspools discharging to those areas received a binary risk score of 100, while all others were scored as 0.
- For **Coastal ocean circulation** proxy. Long-term mean wave power from SWAN model outputs was used as a proxy for coastal exposure and residence conditions at the analysis scale. Raster values were mapped to the coastline grid and scored using a threshold-decay function with a threshold of 1.59 kW m^{-1} to capture the most sheltered decile as highest priority. Although wave power does not capture all circulation processes, it provides a consistent, statewide indicator of relative mixing and exposure for prioritization (Fig. 10).

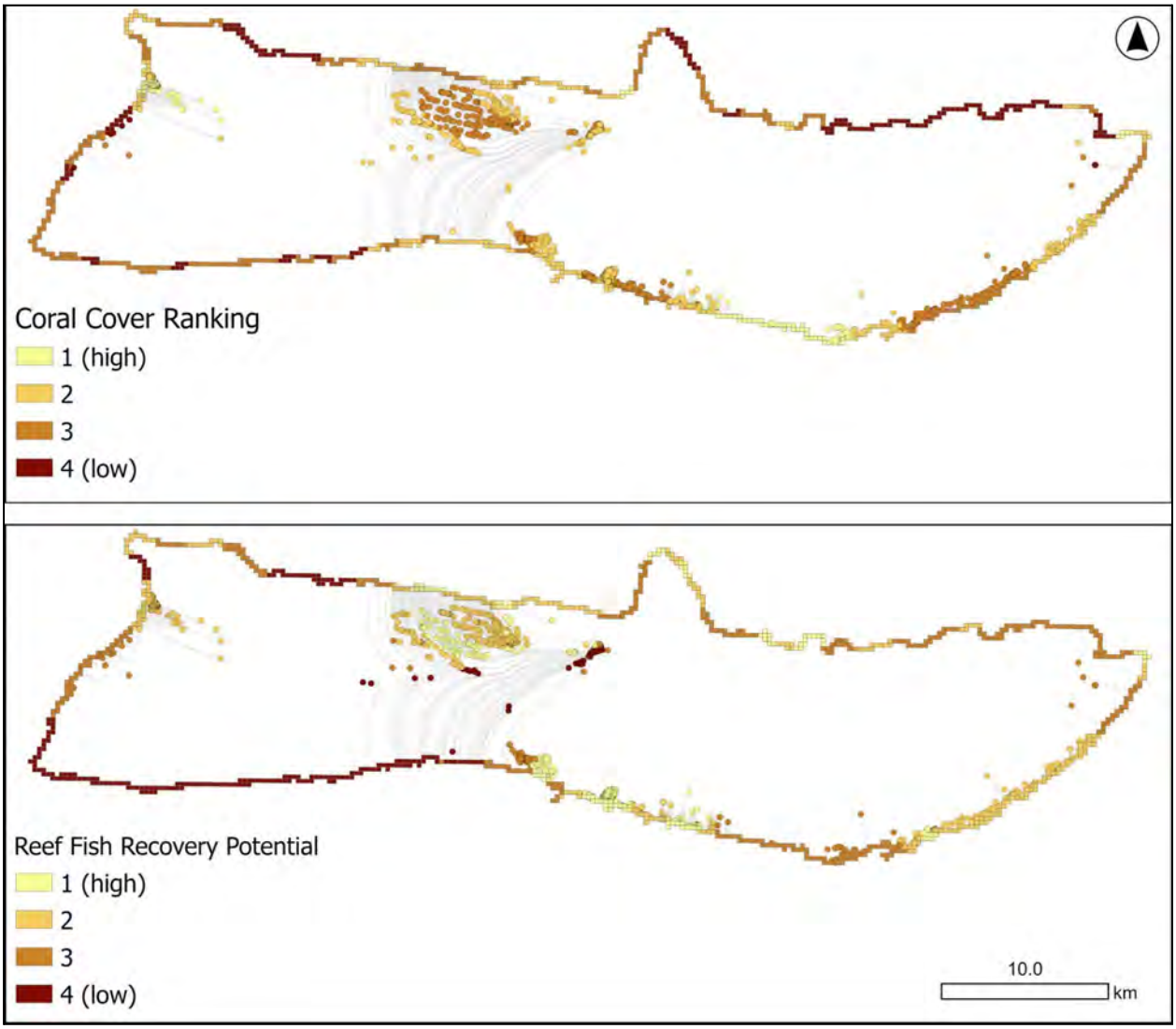


Figure 9. *Ecological risk factor layers for Moloka'i. Top: coral cover ranking, with nearshore areas classified from high (1) to low (4) coral cover. Bottom: reef fish recovery potential, with nearshore areas ranked from high (1) to low (4) based on ecological resilience and recovery value. Cesspool locations are shown in relation to each layer, highlighting where effluent discharge overlaps with sensitive reef ecosystems.*

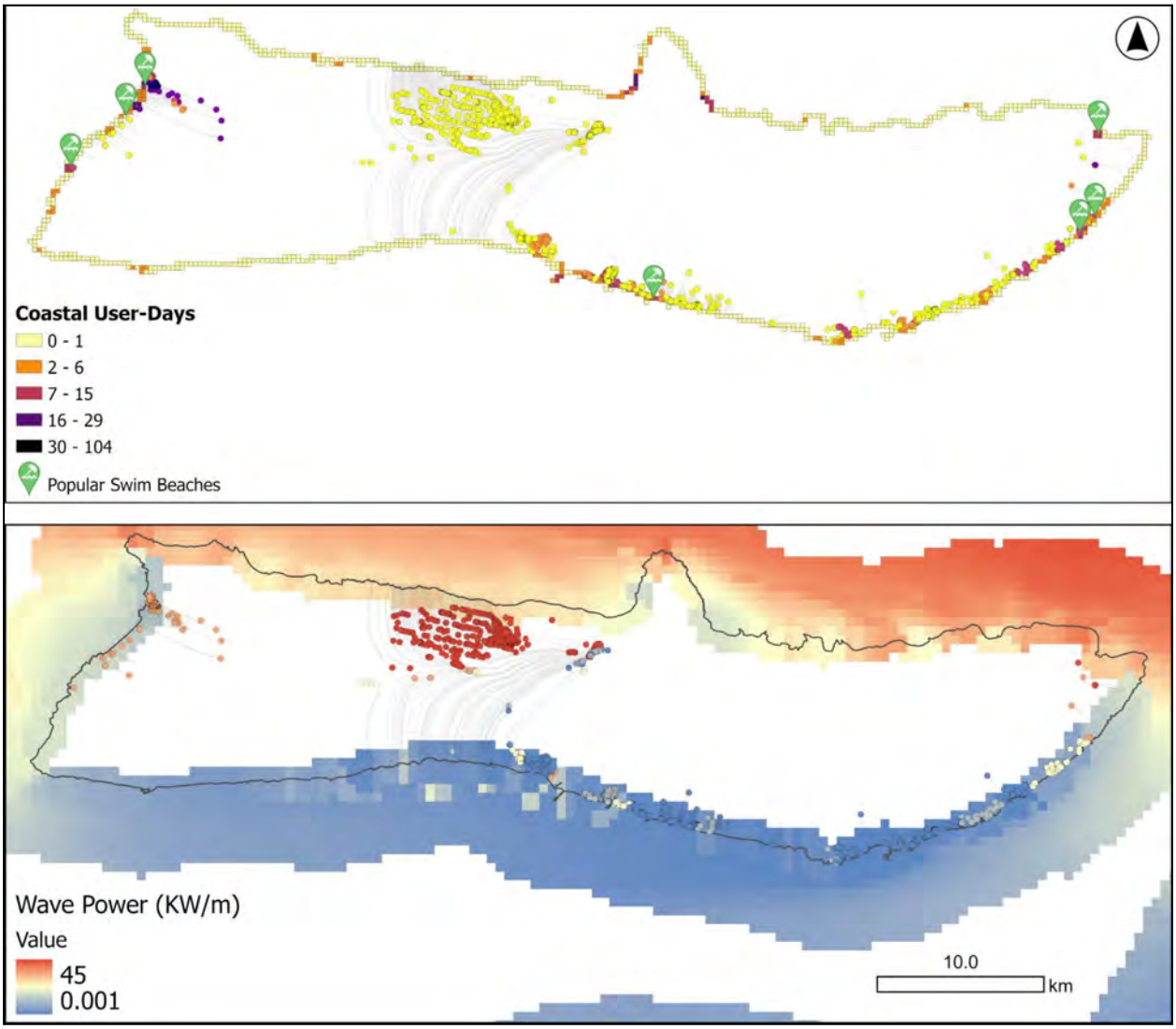


Figure 10. *Recreational and coastal exposure risk factors for Moloka'i. Top: coastal user-days, showing relative recreational intensity from low (0–1) to high (30–104). Separately, the seven popular swim beaches identified for the project are indicated by green markers. Bottom: modeled nearshore wave power (kW/m), with higher values (red) representing greater wave power and thus greater potential for effluent dispersal in the coastal zone, which translates to a lower risk of wastewater impacts in these areas.*

3. Results

This analysis used the most recent statewide cesspool inventory to identify and prioritize cesspools located within the island of Molokaʻi. A total of 1,280 cesspools were identified within Molokaʻi's boundaries, based on available spatial datasets. Because Molokaʻi has relatively few census tracts, block groups, and blocks compared to other islands, and cesspools are distributed in smaller, more localized clusters, the HCPT thresholds were adjusted to be specific to Molokaʻi island. The standard thresholds of 25 cesspools per tract, 20 per block group, and 10 per block were reduced to 25 per tract, 10 per block group, and 5 per block to ensure that smaller but still significant concentrations were captured in the prioritization process. These revised thresholds were applied to rank census tracts, block groups, and blocks by cesspool density, supporting a more accurate identification of high-priority areas for conversion on Molokaʻi. Given the inherent subjectivity in defining thresholds, adoption of the Molokaʻi thresholds in future iterations of the HCPT is recommended, as they provide the broadest applicability across all islands in the Hawaiian chain. All results are publicly accessible in the project repository located on GitHub at https://github.com/cshuler/HCPT_Molokai/tree/main/Scripts/Outputs.

3.1 Tracts

Molokaʻi is divided into only three census tracts: East Molokaʻi (tract 318), West Molokaʻi (tract 295), and Kalawao (tract 319), which covers the Kalaupapa Peninsula (Table 1). The cesspool inventory revealed that Kalawao contained no cesspools in this dataset, reflecting its small population and unique land use. As a result, only East and West Molokaʻi were evaluated for prioritization under the adjusted HCPT thresholds. Applying the prioritization process, the East Molokaʻi census tract was ranked as Priority 1, reflecting the highest cesspool density and potential impacts within the island. Census tract West Molokaʻi was ranked as Priority 3, indicating a comparatively lower density and impact. These results highlight the concentration of coastal cesspools in the eastern portion of the island, which ultimately increases their impacts on valued resources. Figure 11 illustrates the spatial distribution of census tracts and their assigned priority categories.

Table 1. *Census Tract–level cesspool inventory and HCPT prioritization results for Molokaʻi.*

Location	Number of Cesspools	HCPT Priority
East Molokaʻi	722	1 (High)
West Molokaʻi	560	3 (Low)

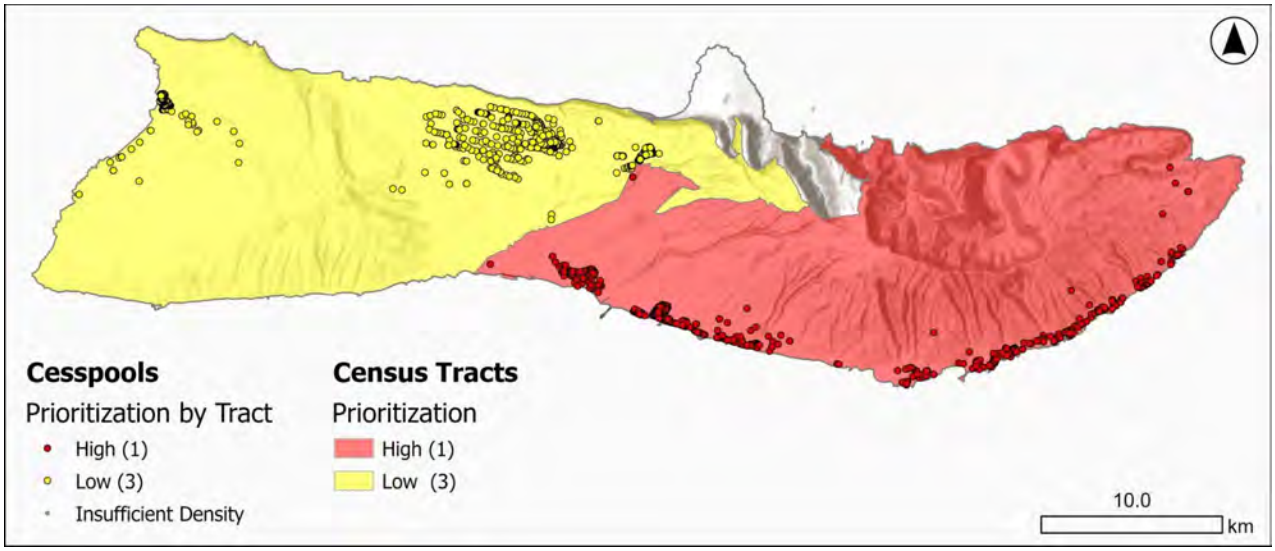


Figure 11. *Census Tract-level cesspool prioritization results for Molokaʻi. Individual cesspools are shown with colors matching their tract priority category.*

3.2 Block Groups

Molokaʻi contains a total of eight census block groups, which provide a finer-scale resolution of population and infrastructure patterns compared to census tracts (Table 2). Applying the adjusted HCPT threshold of 10 or more cesspools per block group, six of the eight block groups on the island met the inclusion criteria for prioritization. The remaining two block groups fell below the threshold and were excluded from further analysis. Among the six qualifying block groups, two were ranked as **Priority 1**, representing the highest relative cesspool density and associated risk, with a combined total of 230 cesspools. One block group was ranked as **Priority 2**, containing 254 cesspools. The remaining three block groups were assigned a **Priority 3** ranking, collectively accounting for 796 cesspools. These Priority 3 areas, while lower in density relative to the rest, still represent significant sources of nutrient and pollutant loading to groundwater and coastal waters due to their overall cesspool counts. The distribution of block group priorities emphasizes that high-priority areas are not limited to a single part of the island but rather occur in multiple distinct locations. Figure 12 displays the spatial layout of Molokaʻi's census block groups and their respective HCPT priority rankings.

Table 2. *Census Block Group–level cesspool inventory and HCPT prioritization results for Molokaʻi.*

Priority Category	Number of Block Groups	Total Cesspools
Priority 1 (High)	2	230
Priority 2	1	254
Priority 3 (Low)	3	796

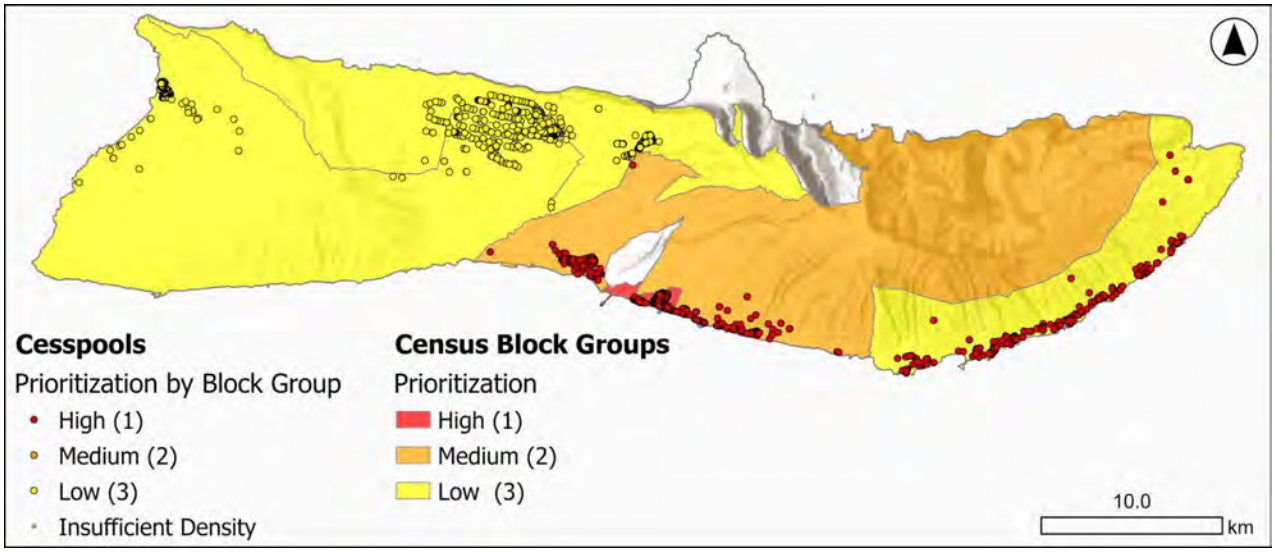


Figure 12. *Census Block Group-level cesspool prioritization results for Molokaʻi. Individual cesspools are shown with colors matching their Block Group priority category.*

3.3 Blocks

Molokaʻi is divided into 781 census blocks, the smallest geographic unit used by the U.S. Census Bureau for spatial data reporting (Table 3). Of these, only 47 blocks contained more than the adjusted HCPT threshold of 5 cesspools, qualifying them for prioritization. An additional 109 blocks contained between 1 and 4 cesspools, while 244 cesspools were located outside of any mapped census block and could not be categorized. In total, 156 census blocks on Molokaʻi contained at least one cesspool. Among the 47 qualifying blocks, 12 were classified as **Priority 1**, containing a total of 238 cesspools. Another 12 blocks were ranked as **Priority 2**, containing 348 cesspools, while 23 blocks were ranked as **Priority 3**,

containing 450 cesspools. This distribution highlights that even within the smallest census units, cesspools are unevenly distributed, with certain clusters exhibiting substantially higher densities and thus greater relative priority for replacement. Figure 13 shows the spatial pattern of prioritized blocks across the island.

Table 3. *Census Block–level cesspool inventory and HCPT prioritization results for Moloka'i.*

Priority Category	Number of Blocks	Total Cesspools
Priority 1 (High)	12	238
Priority 2	12	348
Priority 3 (Low)	23	450

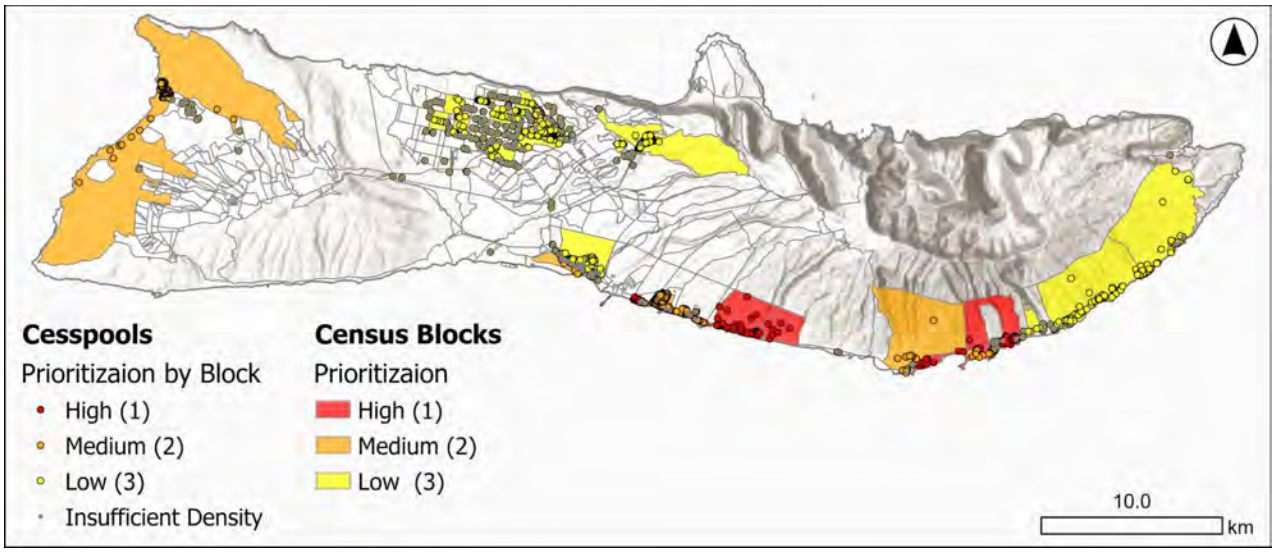


Figure 13. *Census Block-level cesspool prioritization results for Moloka'i. Individual cesspools are shown with colors matching their Block priority category.*

3.4 Breakdown of Individual Risk-Factor Scores by Census Areas

Each cesspool receives an individual score for each risk-factor. These are totalized and weighted to create the individual final prioritization score for each cesspool, which is the metric that is averaged (using the arithmetic mean) across all cesspools in each census area (e.g. tract or block) to define the census area's HCPT Prioritization score. Significant insight can be gained through examining the combined individual scores for each risk-factor

summarized by census area groupings. This analysis yields information about which of the 15 risk factors are predominant in driving the prioritization within each census area.

3.4.1 Tract level

The tract-level results (Fig. 14) show a clear contrast between East and West Molokaʻi, with East Molokaʻi exhibiting consistently higher scores across most risk factors, leading to a much higher final priority score (99.9 compared to 57.8 in the west). In particular, East Molokaʻi shows elevated values for depth to groundwater (171.2), reef fishery priority (240.1), wave power (176.3), cesspool density (148.2), and proximity to domestic wells (110.6), all of which strongly drive its prioritization. East also has markedly higher scores for rainfall, distance to coast, and municipal well proximity, reinforcing its greater overall vulnerability. In contrast, West Molokaʻi shows relatively lower scores across most categories but has localized risk drivers, including higher scores for swim beaches (29.3 vs. 6.4), coastline usage (7.9 vs. 4.6), and well capture zones (8.6 vs. 1.5), indicating specific recreational and drinking water concerns. These patterns highlight that while East Molokaʻi faces broader and more severe cesspool-related risks, West Molokaʻi's risks are tied more narrowly to recreational exposure and well capture sensitivity.

When looking across both tracts together, the most influential factors in driving prioritization are clear: reef fishery and coral reef priorities consistently rank among the highest scores overall, underscoring the importance of ecological impacts. Depth to groundwater, wave power, cesspool density, and proximity to domestic wells also emerge as major risk drivers, reflecting the combined weight of groundwater vulnerability, hydrodynamic exposure, and the distribution of cesspools themselves. Rainfall, coastal proximity, and municipal well proximity further add to the cumulative risks, especially in East Molokaʻi, highlighting the multiple pathways by which cesspool effluent can affect water quality and ecosystems. By contrast, several factors exert relatively minor influence across both regions, including well capture zones (1.5 east, 8.6 west), coastline usage (4.6 east, 7.9 west), and swim beaches (6.4 east, 29.3 west), as well as sea level rise scores (28.1 east, 2.5 west). These consistently lower values indicate that recreational exposure and sea level rise play a limited role compared to the much stronger ecological and hydrological pressures shaping prioritization. Together, the results show that cesspool risk on Molokaʻi is overwhelmingly defined by a small set of dominant ecological and groundwater-related drivers, while other factors make only marginal contributions.

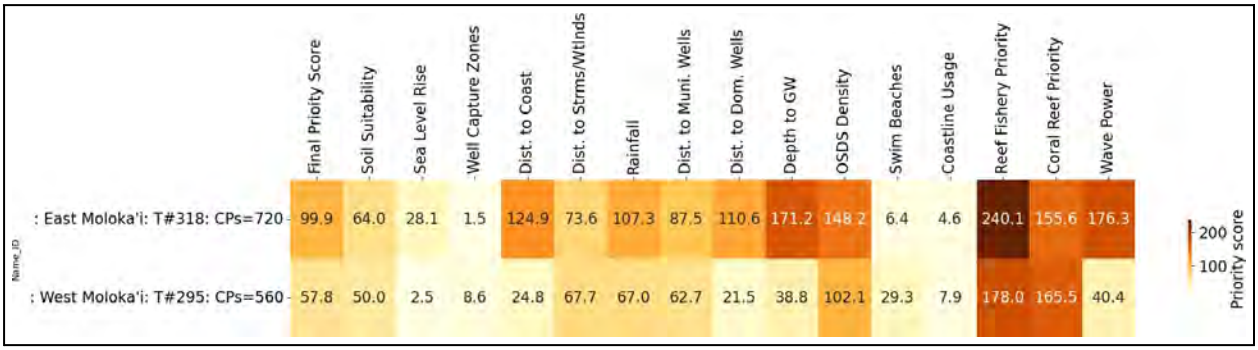


Figure 14. *Pivot table showing census tract–level risk factor scores for Moloka‘i cesspools. Each row represents a census tract, and each column corresponds to one of the 15 HCPT risk factors, except for the first column, which shows the final priority score. Values represent mean scores for all cesspools within each tract, with darker shading indicating higher relative contributions to overall prioritization.*

3.4.2 Block Group Level

Moloka‘i has only eight block groups, six of which contain cesspools, and they vary considerably in size, density, and development. Some block-groups, such as 788 and 789 around Kaunakakai, reflect a more urbanized character, while others are more dispersed or cut across rural clusters. Generally, block groups along the more developed central and eastern parts of the southern coast carry higher composite scores because they intersect more risk factors (Fig. 15).

Of the two **High Priority (1)** block groups, BG 788 and BG 789 together define the island’s highest-risk cluster around Kaunakakai. BG 788, though it contains only 39 cesspools, rises to the top because its cesspools sit in the most sensitive physical setting: extremely shallow groundwater (310), direct connection to high-value reef fisheries (300), and energetic nearshore wave exposure (198). In other words, a relatively small number of systems have a disproportionately large impact because of their location. By contrast, BG 789’s risk is less about setting and more about sheer numbers — with 191 cesspools concentrated at very high density (334), compounded by heavy rainfall recharge and proximity to domestic wells. Taken together, these two groups illustrate the dual pathways by which cesspools create islandwide risk: a few cesspools in very vulnerable environments (BG 788), or a large number of cesspools exerting cumulative stress in a more moderately vulnerable setting (BG 789). The convergence of these pathways in Kaunakakai highlights the town as the focal point of cesspool-driven impacts on Moloka‘i’s coastal waters and water supply.

The two **Medium Priority (2)** block groups, BG 787 (Hoʻolehua/Kualapuʻu, 236 cesspools) and BG 845 (south shore, 254 cesspools), highlight distinct risk pathways. BG 787's score stems from inland drivers, with shallow groundwater (194) and proximity to domestic wells (181) making drinking-water contamination the dominant concern. BG 845, by contrast, represents a coastal cluster where high reef fishery priority (236) and shallow groundwater (166) drive ecological risks, with additional influence from rainfall and municipal well proximity. Together, they show how inland cesspools on Molokaʻi may impact groundwater supplies more heavily while coastal clusters amplify stress on nearshore reefs. The two **Low Priority (3)** block groups, BG 779 (Kepuhi area, 216 cesspools) and BG 829 (upland east, 344 cesspools), represent very different low-risk settings. BG 779's score reflects its coastal position, where high reef fishery (206) and coral reef priority (198) are offset by very low wave power (27) and modest cesspool density, limiting overall risk. BG 829, the island's largest block group, has dispersed upland cesspools with little shallow groundwater influence (12) and only moderate reef/coral scores, resulting in low composite risk despite its large cesspool count.

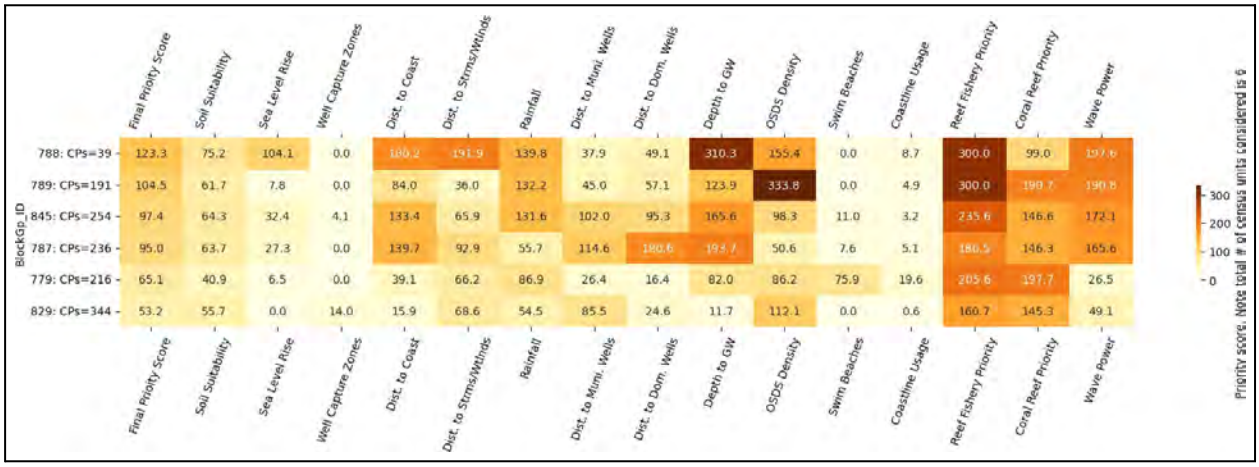


Figure 15. *Pivot table showing census block group–level risk factor scores for Molokaʻi cesspools. Each row represents a block group, and each column corresponds to one of the 15 HCPT risk factors, except for the first column, which shows the final priority score.*

3.4.3 Block Level

At the block level and taken all together, an analysis of the individual risk scores reveal that cesspool impacts on Molokaʻi are strongly shaped by a fairly consistent set of drivers rather than evenly distributed across all of the factors (Fig. 16). The most influential factors that repeatedly show up with the highest intensities across Molokaʻi are reef fishery and coral reef priorities, followed by depth to groundwater and cesspool density. In certain more inland settings, proximity to domestic wells emerges as a concern, while along the coast, wave power and distance to streams provide additional nuance. Factors like soil suitability and slope appear less determinative, showing lower and more uniform values across blocks. Overall, the block-scale data emphasize that cesspool risks across the island concentrate where density overlaps with shallow groundwater or high reef resource value, sharpening local hotspots within broader census areas.

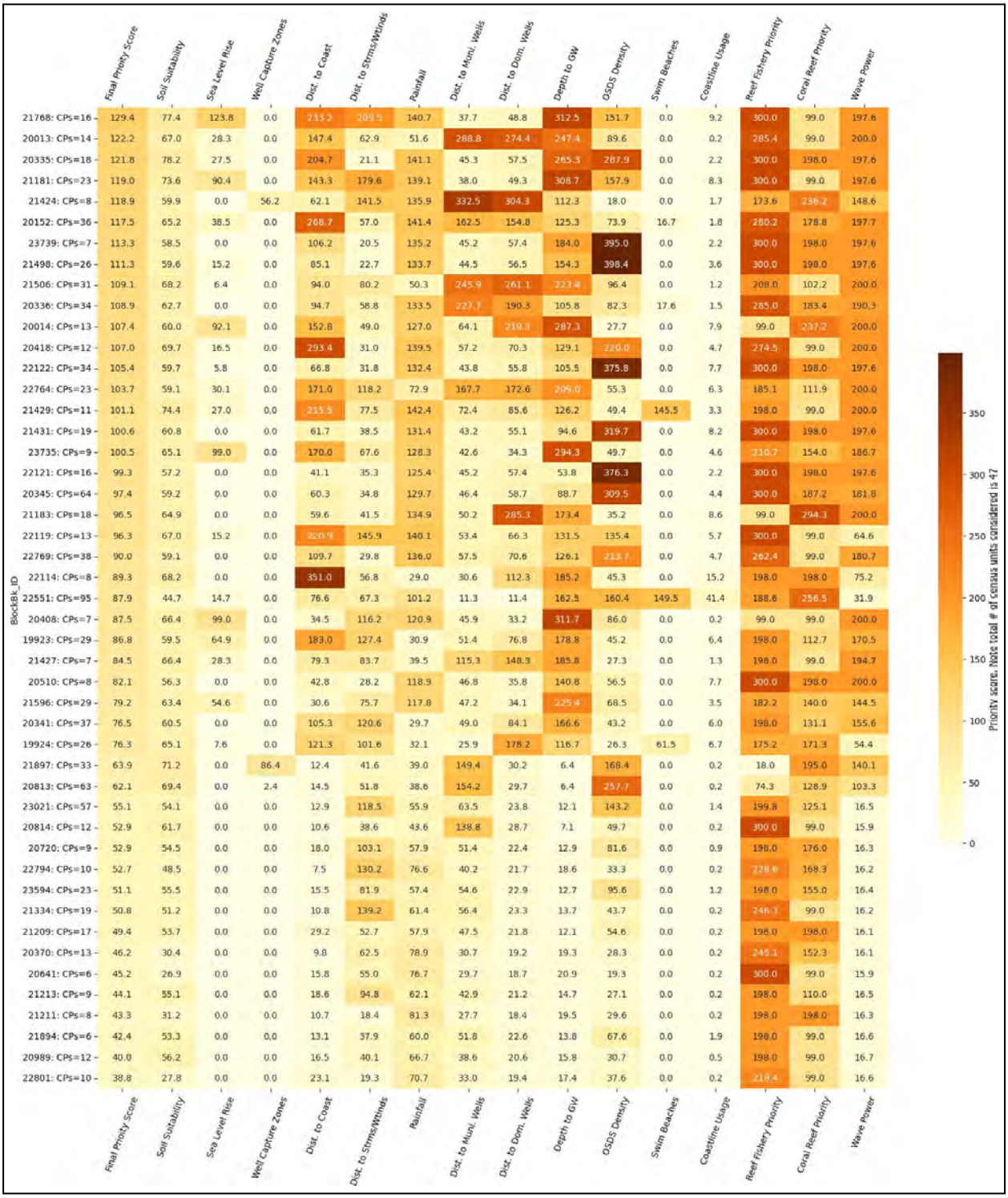


Figure 16. *Pivot table showing census block–level risk factor scores for Moloka'i cesspools. Each row represents a block group, and each column corresponds to one of the 15 HCPT risk factors, except for the first column, which shows the final priority score.*

4. Management discussion

Beyond just providing a prioritization schema to areas across Molokaʻi, the HCPT gives a richness of information regarding the nuanced impacts of each individual cesspool across the diverse array of factors considered. This level of detail enables managers to better target resources and interventions where they will have the greatest impact.

4.1 Resolution-Based Prioritization and Planning Implications for Management

The HCPT produces results at multiple geographic resolutions, each suited to a different management purpose. Census tracts provide a broad, island-wide perspective that is useful for statewide prioritization and funding allocations. Block groups offer a more practical scale for island-level planning and infrastructure targets, while census blocks provide the most localized detail, supporting site-specific assessments, community outreach, and engineering studies. However, census boundaries do not efficiently reflect how cesspools are actually distributed on Molokaʻi. For management purposes it may be useful to also consider various representative cesspool settings, or clusters, that share similar geographies and risk-factor interactions across the island (Fig. 17), namely: 1) high-density residential developments flanking Kaunakakai, 2) low-density but coastal cesspools along the island's southern shoreline, 3) elevated, dispersed agricultural cesspools in Hoʻolehua and Kualapuʻu and 4) resort-area cesspools in Kepuhi Beach. Although these settings do not align perfectly with any of the census area resolutions, they are somewhat captured at the block group level, which makes block groups perhaps the most practical framework for interpreting Molokaʻi's cesspool landscape.

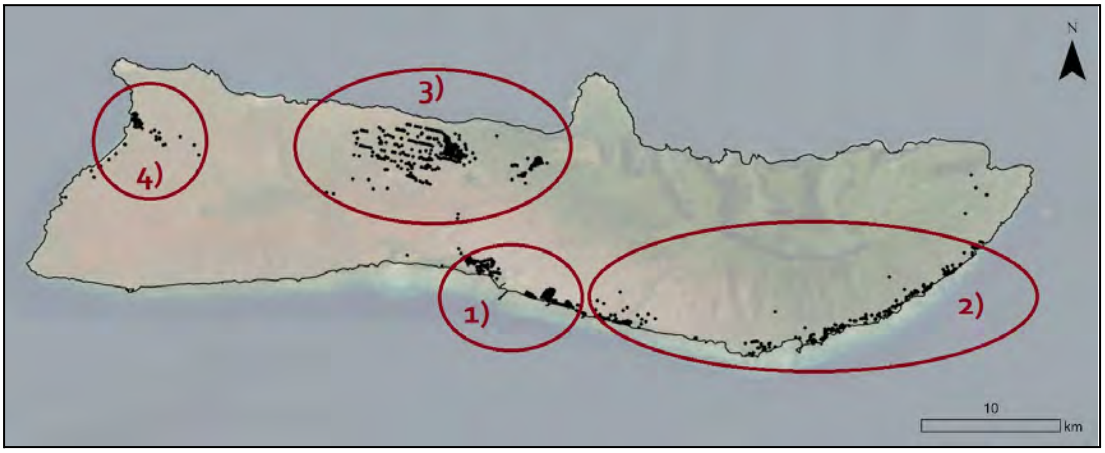


Figure 17. *Cesspool distribution on Molokaʻi grouped into four representative settings that provide a practical perspective for discussion outside of the Census area framework: (1) Kaunakakai, (2) southern shoreline, (3) Hoʻolehua/Kualapuʻu uplands, and (4) Kepuhi Beach.*

Overall the HCPT results suggest that the developments near Kaunakakai emerge as the most logical target for sewer expansion, given the density of cesspools, their proximity to the existing County-operated Kaunakakai Wastewater Reclamation Facility, and the clear Priority 1 status in all Census area resolutions. By contrast, cesspools along the southern coast present strong ecological and drinking water risks, particularly to coastal ecosystems and nearby wells, but their low density makes them less suitable for centralized sewering. Instead, distributed upgrades such as advanced onsite treatment systems may be a more effective solution. The upland areas of Hoʻolehua and Kualapuʻu, characterized by dispersed, elevated, and primarily agricultural cesspools, may pose lower ecological risk, but replacement strategies need to account for cost-effectiveness and the rural character of these communities. The Kepuhi Beach area represents a different situation, where the identified cesspools are tied to resort development and could be addressed through targeted upgrades or connections to private systems. However, prioritization here is complicated by uncertainty regarding the Kaluakoʻi Wastewater Treatment Plant service area; while the facility is documented in PUC records, service boundaries are unclear, and many properties remain listed as cesspools in the available inventory. This highlights important data gaps that must be addressed before detailed planning can proceed. Taken together, this assessment demonstrates that the block group resolution provides the "sweet spot" for planning on Molokaʻi, offering enough granularity to capture distinct cesspool settings while remaining interpretable for management decisions.

4.2. Key Risk factors and Unique Prioritization Characteristics of Molokaʻi

Molokaʻi's cesspool landscape is defined less by sheer numbers than by its rural and dispersed character. Along the southern shoreline, cesspools are scattered across low-density communities that are often lower-income and more rural than comparable coastal areas on other islands. Despite the relatively small population, these areas overlap with highly sensitive reef and fishery resources, meaning that even modest effluent loads may be generating disproportionate ecological harm. What stands out in the Molokaʻi HCPT results is the consistency of the dominant risk drivers across settings: reef fishery and coral priorities, as well as cesspool density, shallow groundwater, and wave power repeatedly emerge as the most important factors. This uniformity across census units suggests that management strategies may be most effective if focused on these few high-leverage drivers rather than attempting to address a broad spread of minor ones. Specifically, this means management targets could be at the intersection of where high value ecological assets (reefs and fish) overlap with low wave power (minimal dilution) and shallow groundwater (rapid transport).

At the same time, Moloka‘i faces unique management challenges that differ from its neighbors. In the upland agricultural communities of Ho‘olehua and Kualapu‘u, cesspools are more dispersed and located at higher elevations, creating different risk profiles and raising questions about cost-effectiveness and appropriate technologies for replacement. In contrast, the un-sewered neighborhoods located near Kaunakakai present the island’s only clear opportunity for sewer expansion, given their density, proximity to the County’s wastewater treatment facility, and role as the island’s central hub. Unlike larger islands with multiple towns and treatment plants, Moloka‘i has little redundancy in wastewater infrastructure, meaning that solutions must balance targeted sewer expansion with distributed onsite upgrades. Finally, the island’s limited number of census tracts and block groups constrains planning scale, with fewer units available to smooth out variation. This makes Moloka‘i’s prioritization results both sharper and more sensitive to outliers, underscoring the importance of careful interpretation in translating them into management action.

5. Conclusions and Recommendations

Moloka‘i, an island of approximately 7,400 people, has 1,280 documented cesspools, meaning that over 60 percent of its approximately 2,100 households still depend on cesspools for wastewater disposal. This project applies the Hawai‘i Cesspool Prioritization Tool (HCPT) framework to these 1,280 cesspools, generating the first cesspool prioritization schema for the island of Moloka‘i. The analysis integrates statewide datasets with Moloka‘i-specific updates, producing risk scores at multiple resolutions to capture both island-wide patterns and localized impacts. Results indicate that a small set of ecological and hydrologic factors consistently drive prioritization, underscoring the extent to which Moloka‘i’s cesspools place pressure on nearshore ecosystems and groundwater resources despite the island’s rural character. All input data, code, and outputs are publicly accessible at https://github.com/cshuler/HCPT_Molokai.

5.1 Management Recommendations

- **Focus on Moloka‘i’s Unique Risk Drivers**
  Management efforts should directly target the consistent dominant drivers that appear across settings: reef fishery and coral reef priorities, cesspool density, shallow groundwater, and wave power. These factors repeatedly shape high-priority outcomes across tracts and block groups, underscoring the need for integrated approaches that address both groundwater vulnerability and nearshore ecological impacts.

- **Opportunities for Sewer Expansion (Kaunakakai)**
  Kaunakakai represents the clearest opportunity for centralized sewer expansion on Molokaʻi. The area has high cesspool density, extremely shallow groundwater, and is adjacent to critical reef ecosystems, while also lying within proximity to existing county wastewater infrastructure. Sewer expansion here would address the island's single largest cluster of cesspools and provide the greatest immediate reduction in risk.

- **Prioritization Framework for Future Action**
  While statewide prioritization has been based on census tracts, Molokaʻi's results show that block groups provide a more practical and interpretable scale for management. Future versions of HCPT could adopt lower thresholds for blocks and block groups (10 per group, 5 per block) to reflect Molokaʻi's smaller and more rural cesspool clusters. This approach will generate a larger number of priority areas, but the refinement is desirable because it distinguishes between settings that would otherwise be lumped together. For example, high-density cesspool areas near Kaunakakai, Kamiloloa, and Kawela likely require different management strategies than the dispersed but still coastal cesspools farther east along the south shore. Using block groups as the default planning scale would allow these differences to be captured and acted upon more effectively.

References

Flanagan, K, B. Dixon, T. Rivenbark, and D. Griffin. (2019). "An Integrative GIS Approach to Analyzing the Impacts of Septic Systems on the Coast of Florida, USA." Physical Geography, September, 1–26. https://doi.org/10.1080/02723646.2019.1671297.

Giambelluca, T.W., Q. Chen, A.G. Frazier, J.P. Price, Y.-L. Chen, P.-S. Chu, J.K. Eischeid, and D.M. Delparte. (2013). Online Rainfall Atlas of Hawai'i. Bull. Amer. Meteor. Soc. 94, 313-316,doi: 10.1175/BAMS-D-11-00228.1.

Hawai'i Climate Change Mitigation and Adaptation Commission. (2021). State of Hawai'i Sea Level Rise Viewer. Version 1.16. Prepared by the Pacific Islands Ocean Observing System (PacIOOS). Office of Conservation and Coastal Lands. http://hawaiisealevelriseviewer.org. Accessed [Aug 2025].

HI-GIS Online - State of Hawaii GIS Program. (2017). SoH_Imagery/Hawaii_DSM (Digital Surface Model for the Hawaiian Islands, 1999–2017) [ArcGIS ImageServer]. Geodata Hawaii Open Data Portal. Retrieved from https://geodata.hawaii.gov/arcgis/rest/services/SoH_Imagery/Hawaii_DSM/ImageServer Accessed [Aug 2025].

HI-GIS Online - Hawai'i Office of Planning and Sustainable Development, Statewide GIS Program. (2020a). Coastline [GIS dataset]. Retrieved from: https://geoportal.hawaii.gov/datasets/cchnl::coastline Accessed [Aug 2025].

HI-GIS Online - Hawai'i Office of Planning and Sustainable Development, Statewide GIS Program. (2020b). Wetlands and Emergent Ponds [GIS dataset]. Retrieved from: https://geoportal.hawaii.gov/datasets/cchnl::wetlands/ Accessed [Aug 2025].

HI-GIS Online - Hawai'i Office of Planning and Sustainable Development, Statewide GIS Program. (2021). Streams [GIS dataset]. Retrieved from https://geoportal.hawaii.gov/datasets/HiStateGIS::streams/ Accessed [Aug 2025].

Mezzacapo, M and Shuler, C, (2022) Hawai'i Cesspool Hazard Assessment & Prioritization Tool – 2022 Update Report & Technical Appendices. Hawai'i Department of Health, Wastewater Branch retrieved from: https://health.hawaii.gov/wastewater/files/2022/11/prioritizationtoolreport.pdf Accessed [Aug 2025].

Tetra Tech, Inc. and University of Hawai'i Coastal Geology Group. (2017). Sea Level Rise – Exposure Area. https://planning.hawaii.gov/gis/download-gis-data-expanded/. Accessed [Aug 2025].

USGS (2013) USGS 10-m Digital Elevation Model (DEM): Hawaii: Molokai, [gridded dataset]. Pacific Islands Ocean Observing System (PacIOOS) ERDDAP. Retrieved from https://pae-paha.pacioos.hawaii.edu/erddap/griddap/usgs_dem_10m_molokai.html Accessed [Aug 2025].

Whittier, R., Rotzoll, K., Dhal, S., El-Kadi, A.I., Ray, C., & Chang, D. (2010). Groundwater source assessment program for the state of Hawai'i, USA: Methodology and example application, J. Hydrogeology, 18: 711–723.